

INTEGRITY RESEARCH INSTITUTE

To:

From: Thomas Valone, President

Attn.

Steven

- ☐ Urgent
- ☒ For Review
- ☐ Please Comment
- ☐ Please Reply

Date sent:

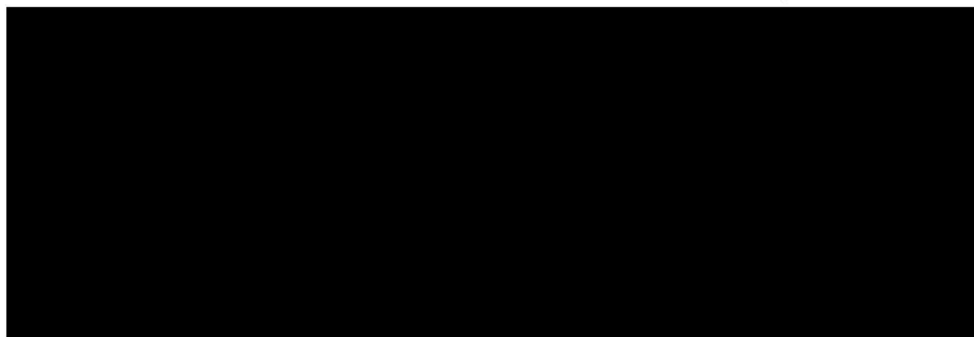
12-5-02

Number of pages including cover page:

5

Message:

Hi



FLUKE

www.specialized.net

Full Specs
on the web!**ScopeMeter® 190 Series****Oscilloscopes for Field Applications**

- Choose 60-, 100 or 200-MHz bandwidth
- Up to 2.5 GS/s real-time sampling
- Automatic capture and replay of 100 screens
- Deep memory for high-resolution ScopeRecord™
- Easy to use with Connect-and-View™ triggering

Speed, performance and analysis power are combined in the ScopeMeter® 190 Series. For more demanding applications, these high-performance oscilloscopes offer specifications usually found only on top-end bench instruments. Models 192, 196 and 199 feature up to 200 MHz bandwidth, up to 2.5 GS/s real-time sampling plus advanced memory and triggering functions. Each rugged instrument is ideal for engineers who need the full capabilities of a high-performance oscilloscope in a handheld, battery-powered instrument. In addition this versatile three-in-one test tool also incorporates a true-rms multimeter and TrendPlot™ paperless recorder.

High sampling rates give you the required resolution for detailed signal analysis. With a maximum real-time sampling rate up to 2.5 GS/s per input, you can see what really happens with 400-ps resolution. Each input has its own digitizer, so you can acquire two waveforms simultaneously and analyze them with the highest resolution and detail. If an anomaly flashes by on the screen, just press the Replay button to see it again.

In normal use, the instrument continuously memorizes the last 100 screens. Each time a new screen is acquired, the oldest is discarded. At any moment, you can freeze the last 100 screens and scroll through picture by picture or replay as live animation. Cursors can be used for further analysis. You can even use the advanced trigger capabilities to capture up to 100 specific events. Two sets of 100 captured screens with individual time stamps can be stored for later recall or download to a PC.

In ScopeRecord™ mode, each ScopeMeter's memory stores 27,500 points per input for high-resolution recording of events up to 30 hours. Yet it still captures fast intermittents and glitches as short as 50 ns. This continu-

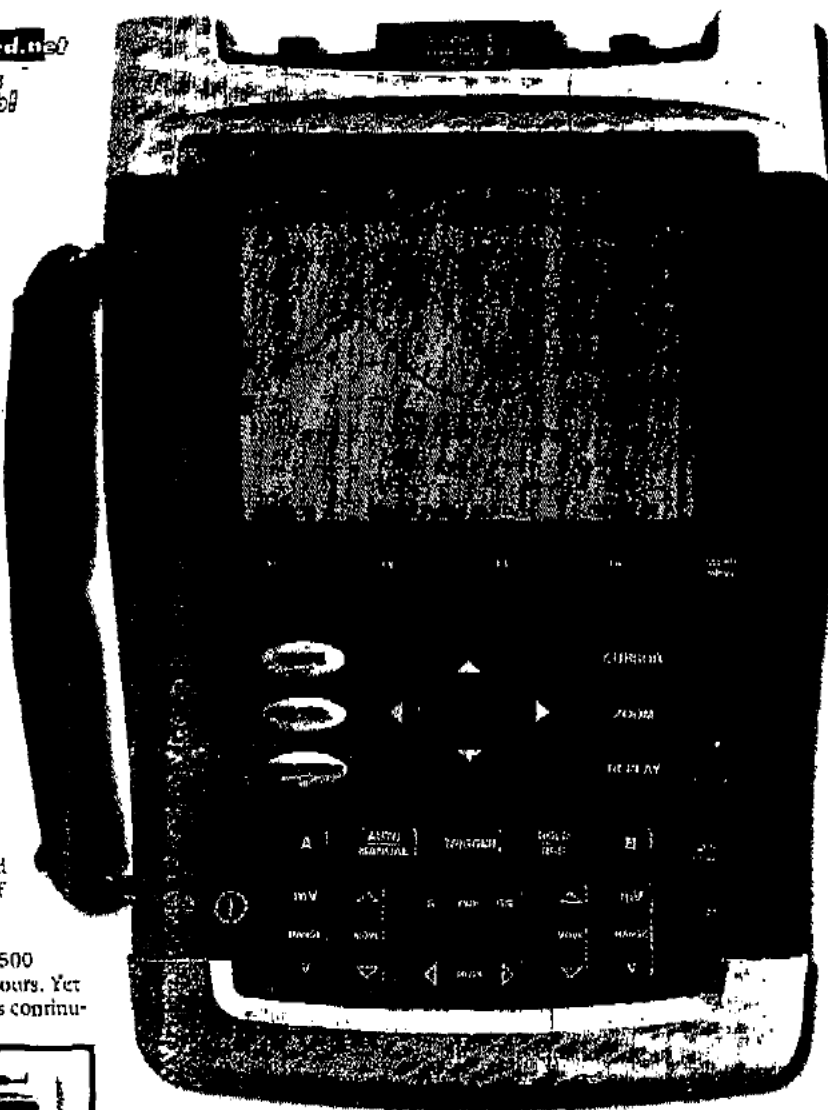
ous roll mode also stores events like motion profiles, UPS, power supply and motor start-ups. And with 100x zoom, you can look at the smallest details like individual power cycles. Two of these 27,500-point recordings can be stored for later analysis.

A wide choice of advanced automatic and manual and external trigger modes gives you the flexibility to capture just about any signal you'll encounter. Connect-and-View™ automatic triggering is ideal for fast and easy checking of multiple test points. It provides a stable, reliable and repeatable display of virtually any signal without touching a button. Manual modes include time delay, video and pulse-width triggering. A fully isolated external trigger input is included for troubleshooting time relationships between two input signals synchronized to a third signal.

For analysis of waveforms, ScopeRecord and TrendPlot recordings, the ScopeMeter 190 Series features 24 automatic waveform measurements plus cursors, zoom and real-time clock. Analysis can be made directly or later when back in the office. Up to two recordings and 10 scope screens can be stored for analysis, printout or download to a PC. In paperless recorder mode, you can plot the minimum and maximum peak values and average over time — up to 16 days. The two inputs can plot any combination of volts, amps, temperature, frequency and phase with time and date stamp, leading you to the cause of those faults quickly. Cursors and zoom help you analyze the captured TrendPlot.

The 190 Series with three independently floating isolated inputs is safety certified for measurements in 1000-V CAT II and 600-V CAT III environments. Multi-purpose scope probes cover a wide measurement range from mV to kV, making these ScopeMeters ideal for a variety of micro electronics to electrical power applications.

The NiMH battery provides four hours of continuous operation and recharges in just four hours. Battery saving and auto power down functions take you easily through the working day. Each ScopeMeter also can be line powered by the included adapter/battery charger.

**Fluke 190 Series Common Specifications**

Inputs and Digitizers	Two plus external/DMM input
Independently Floating Isolated Inputs	Up to 1000V between inputs, references and ground
Input Sensitivity	8.4mV-100V/div
Trigger Types	Connect-and-View™, free run, single shot, edge, delay, video, pulse, pulse width and external
Glitch Capture	50 ns
Cursors and Zoom	✓
Scope Measurements	Seven cursor + 24 automatic
Maximum Record Length	27,500 points per input in ScopeRecord mode 1000 points per input in scope mode
Capture Last 100 Screens	Automatic with replay capability
Dual Input TrendPlot	With cursors and zoom
Memory for Screens/Screen	10 screens and set-ups
Memory for Recordings	Two: each can store 100 scope screens, a ScopeRecord of a TrendPlot
True-rms Multimeter	5000 counts, volts, amps, Ohms, conductivity, diode, temp

OSCILLOSCOPES



Select the Right ScopeMeter for Your Needs

Choose from three dual-input 190 Series models, offering different bandwidths. The 60-MHz 192 has 500 MS/s maximum real-time sampling rate. Model 196 is the 100-MHz version with a maximum real-time sampling rate of 1 GS/s. These two cost-effective models provide advanced functionality without paying for bandwidth you don't need. But when a 200-MHz bandwidth is necessary for capturing high-frequency signals found in today's state-of-the-art electronics, choose the powerful 199 ScopeMeter with 2.5 GS/s maximum real-time sampling rate.

Standard and calibrated (C) ScopeMeters include line voltage adapter, battery pack, probes, probe accessories and multimeter test leads. The calibrated versions maximize meter reliability. Traceable to MIL-STD-45662A and NIST, they add certification document, label and seal. When you order a standard meter with an SCC190E kit, you also get three additional accessories: FlukeView® for Windows® software, optically isolated RS-232 interface cable plus hard-shell carrying case. This software/cable/case kit also may be ordered separately. Each 4.3-pound ScopeMeter measures 10.1" x 6.6" x 2.5". Three-year warranty.

192/003 192 ScopeMeter (60 MHz)

044X272 1-2, \$1990.00 ea.; 3-up, \$1916.00 ea.

192/003 192 Calibrated ScopeMeter (60 MHz)

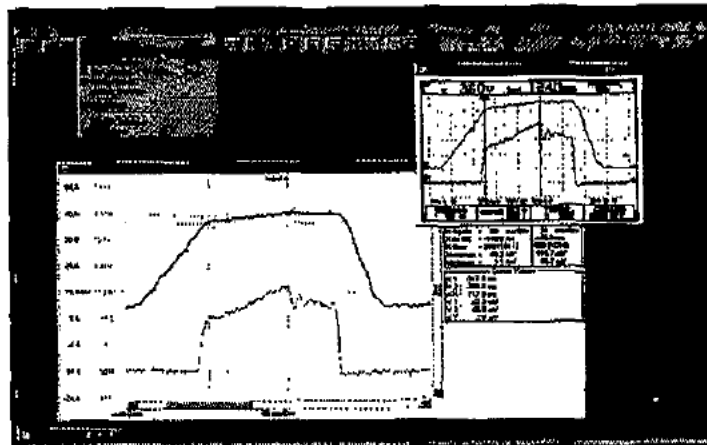
044X272C \$2112.00 ea.

→ 192/003S 192 ScopeMeter (60 MHz) with SCC190E Kit

044X274 1-2, \$2380.00 ea.; 3-up, \$2290.00 ea.

196/003 196 ScopeMeter (100 MHz)

044X280 1-2, \$2290.00 ea.; 3-up, \$2204.00 ea.



SCC190E Kit Adds FlukeView® Software and More

Each ScopeMeter model also is available with an SCC190E accessory kit. This three-piece addition features FlukeView® for Windows® Software to help you get more out of your ScopeMeter. For documenting, it transfers waveform, screen and measurement data from the ScopeMeter to a PC. Then you can print or import the data into your report. To archive, it creates a library of waveforms with your comments for easy reference and comparison. For analysis, use cursors, perform spectrum analysis or export data to other analysis programs. Connect any ScopeMeter to a PC via the included optically isolated RS-232 interface cable. A protective hard-shell carrying case rounds out the SCC190E kit. It provides safe and convenient storage of the instrument with all its accessories.

Fluke 190 Series Specification Comparison

	Fluke 192	Fluke 196	Fluke 199
Bandwidth	60 MHz	100 MHz	200 MHz
Max Real-Time Sample Rate	500 MS/s	1 GS/s	2.5 GS/s
Timebase Range	5 ns-2 ms/div	5 ns-2 ms/div	5 ns-2 ms/div

196/003 196 Calibrated ScopeMeter (100 MHz)

044X280C \$2412.00 ea.

196/003S 196 ScopeMeter (100 MHz) with SCC190E Kit

044X286 1-2, \$2680.00 ea.; 3-up, \$2578.00 ea.

199/003 199 ScopeMeter (200 MHz)

044X293 1-2, \$2590.00 ea.; 3-up, \$2492.00 ea.

199/003 199 Calibrated ScopeMeter (200 MHz)

044X293C \$2712.00 ea.

199/003S 199 ScopeMeter (200 MHz) with SCC190E Kit

044X295 1-2, \$2980.00 ea.; 3-up, \$2866.00 ea.

SCC190EFG Software/Cable/Case Kit for Fluke 190 Series

044X958 \$390.00 ea.

Optional ScopeMeter Accessories

044X735 DP 120 Differential Voltage Probe \$395.00 ea.

045X016 80i-500i 500-A AC Current Probe 170.00 ea.

044X300 80i-110i 100-mA to 100-A AC/DC Current Probe 395.00 ea.

045X955 110005 100-A to 1000-A AC Current Probe 349.00 ea.



Connect-and-View captures the most complex motor drive signals to measure a number of testpoints in quick succession.



High Sampling Rates give you the required resolution for detailed signal analysis to see what's really happening.



Pulse-Width Triggering allows you to find anomalies like missing pulses.



Analysis Power won't let you down. Use ScopeRecord's 27,500 points of memory and zoom in to maximum detail.

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Full Specs
on the web!

MultiLog™ Series Datalogging MultiMeters

- True-rms for ac volts and current
- PC interface with included software

Log and view up to 43,000 data points as fast as 20 per second and generate reports from logged field data. Monitor and analyze electrical parameters.

Make true-rms ac voltage and current measurements. Ten selectable datalogging sample rates range from .05 to 480 seconds. View stored data on LCD with quick Min/Max and trend displays. A 5000-count backlit LCD with bargraph offers high resolution and updates five times per second. Features .08% basic dc V accuracy and °F/°C temperature. Data point memory is 5400 for ML710 and 43,000 for ML720. Includes 10-A CAT III 1000-V test lead set, Type-K thermocouple probe, Windows® 95/98-compatible software, RS-232 cable and 9-V battery.

MultiLog Specifications

Max Voltage (dc/true-rms ac)	1000 V
Max Current (dc/true-rms ac)	10 A
Max Ohms	50 MΩ
Max Capacitance	9999 µF

160X109 ML710 MultiLog DMM	1-5, \$249.00 ea.; 6-up, \$230.00 ea.
160X110 ML720 MultiLog DMM	1-5, 299.00 ea.; 6-up, 276.00 ea.
160X391 CA899 Carrying Case	15.00 ea.

MultiPro™ Series High-Performance Autoranging DMMs with PC Interface

- High basic dc V accuracy of 0.08%
- Double-injection CAT III 1000-V test leads

MultiPro™ DMMs offer .08% basic dc voltage accuracy, and backlit 6000-count LCD provides high resolution. Measures ac/dc V and A, resistance, frequency, capacitance, diode and continuity. Features data hold and low-resistance auto-lead zero plus smart auto power off which is disabled if signal is present at test leads.

Wide ac voltage bandwidth is 40 Hz-20 kHz for audio frequency plus digital electronic production test and repair. Optional RS-232 PC interface with Windows® 95/98-compatible software lets user collect, display, plot, save or export data or graphs. Use true-rms MP520 and MP530 for V and A measurements of distorted waveforms in office and industrial locations.

Model MP510 does all the basics. MP520 adds true-rms ac V and A. The MP530 also has °F/°C temperature, relative, min, max, max-min, 5-ms peak hold and analog bargraph for 5X resolution to detect signals with 60 updates a second. Includes stand, 10-A double CAT III 1000-V test leads, 9-V battery and holster. Model 530 adds temperature probe.

MultiPro Specifications

Max Voltage (dc/true-rms ac)	1000 V
Max Current (dc/true-rms ac)	10 A
Max Ohms	50 MΩ
Max Capacitance	9999 µF

160X965 MP510 Basic MultiPro DMM	1-5, \$139.00 ea.; 6-up, \$128.00 ea.
160X970 MP520 True-RMS MultiPro	1-5, 159.00 ea.; 6-up, 147.00 ea.
160X973 MP530 Full-Featured DMM	1-5, 179.00 ea.; 6-up, 165.00 ea.
160X984 SW800 MultiPro Software	49.00 ea.

Mini Digital Multimeter Ideal Choice for a Basic DMM

- Large 3.5-digit LCD with 1999 maximum reading
- Sampling rate of 2.5 times per second

This compact DMM is rugged enough for field use. Use rotary switch to select seven functions: dc/ac V, dc A, resistance, diode, transistor and

380602 Specifications

MultiTec™ Series

Autoranging DMMs with RS-232 PC Interface

- Extra-large display with unique prompts
- Data hold and relative functions

CAT II MultiTec™ DMMs have a unique display with prompts indicating correct test probe connections, blown fuses and to disconnect power for passive functions. High-contrast 4000-count display is oversized for easier viewing. Features data hold and relative functions.

Extra-large display, connection and voltage warnings plus PC interface are ideal for training and technical schools. Capacitance and diode functions are good for appliance installation and repair. Use dc functions. duty cycle and capacitance for automotive and marine applications. Optional PC interface and Windows® 95/98-compatible software lets user collect, display, plot and save data.

MultiTec Specifications

Max Voltage (dc)	1000 V
Max Voltage (ac)	700 V
Max Current (dc/ac)	20 A
Max Ohms	40 MΩ
Max Capacitance	100 µF

Basic dc V accuracy is 0.5% for MT310, 0.3% for MT320 with °F/°C temperature and 0.1% for backlit MT330 with °F/°C temperature, true-rms ac V and current. Includes built-in stand, test leads, 9-V battery and holster. MT320 and MT330 add temperature probe.

Basic dc V accuracy is 0.5% for MT310, 0.3% for MT320 with °F/°C temperature and 0.1% for backlit MT330 with °F/°C temperature, true-rms ac V and current. Includes built-in stand, test leads, 9-V battery and holster. MT320 and MT330 add temperature probe.

160X976 MT310 Basic MultiTec DMM	1-5, \$ 79.00 ea.; 6-up, \$ 73.00 ea.
160X979 MT320 DMM w/ Temperature	1-5, 99.00 ea.; 6-up, 94.50 ea.
160X980 MT330 Full-Featured DMM	1-5, 129.00 ea.; 6-up, 119.00 ea.
160X988 SW805 MultiTec Software	29.00 ea.

MultiMaster™ Series

Precision True-RMS MultiMeters

- 500,000/50,000-count backlit LCD
- Benchtop accuracy with high resolution

Count on MultiMaster™ meters to provide 0.02% basic dc V accuracy with 1-µV max resolution at 50,000 counts. Precision true-rms tools feature 50,000-count backlit LCD with bargraph (500,000 counts for dc V and Hz).

MultiMaster Specs

Max Voltage (dc/true-rms ac)	1000 V
Max Current (dc/true-rms ac)	10 A
Max Ohms	50 MΩ
Max Capacitance	9999 µF

For process current measurements, 4-20 mA is displayed as 0-100%. Frequency is measured to 200 kHz for high voltage and to 2 MHz for signal level applications with adjustable sensitivity. Includes Max, Min and Max-Min recording plus dBm measurements with 20 selectable impedances. Functions include capacitance, resistance, ac+dc V, continuity, duty cycle and diode tests. Expect high resolution of 0.01 Ω, 0.01 µA and 0.1°.

Peak C captures transient pulses. Includes relative and hold. MM560 has 40-Hz to 20-kHz true-rms ac V bandwidth. Enhanced 20 Hz to 100 kHz MM570 with temperature includes T1, T2 and T1-T2 dual Type-K °F/°C temperature. Each includes rubber holster, test leads, K thermocouple probe and 9-V battery.

160X955 MM560 MultiMaster DMM	1-5, \$229.00 ea.; 6-up, \$211.00 ea.
160X960 MM570 DMM with Temperature	1-5, 299.00 ea.; 6-up, 276.00 ea.
160X982 SW810 MultiMaster Software	79.00 ea.

Pen DMM with Logic Test

Versatile Troubleshooter

- Eight functions
- Tests digital circuitry

Field service tool troubleshoots logic, ac/dc circuits, R & D and circuit boards

EXTech

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audible continuity. Includes stand, hanger, leads, spare fuse and 9-V battery.

Measurement	Range	Measurement Range
Voltage (ac)	4	2-1000 V
Voltage (dc)	2	200-750 V
Current (ac)	3	2-200 mA
Resistance (Ω)	6	200-20 MΩ

160X108 380602 Mini DMM 1-5, \$39.00 ea.; 6-up, \$36.00 ea.

160X382 409992 Optional Vinyl Pouch Carrying Case 10.00 ea.

plus tests digital circuitry. Measures 300 mV-500 V dc, 3-500 V ac, ac/dc current to 300 mA and resistance to 30 MΩ. Large, 3200-count, high-contrast LCD has bargraph. Autoranging DMM offers eight functions including data hold and range hold. Features diode test with forward voltage drop and continuity test with audible beep <20 Ω. For logic test, high/low LED and LCD indicate CMOS/TTL threshold level. Has overload indication. Includes two 1.5-V AAA batteries, extra fuses and test lead with alligator clip.

160X196 381626 Pen DMM with Logic Test 1-5, \$49.00 ea.; 6-up, \$46.00 ea.

NEXT-DAY DELIVERY ONLY \$19.95!
<http://www.specialized.net>



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Full Specs
on the web!

Economical Service Meter

Large, Easy-to-Read Display

- Auto-off function
- 32-segment bargraph

Compact, affordable meter measures dc voltage to 1000 V, ac voltage to 750 V, fused dc current to 200 mA and resistance to 2 MΩ. Tough, dependable Wavetek® Meterman™ field service meter with 3.5 digits also features high-speed continuity beeper and diode testing. Read large .67" high numerals easily from a distance. Incorrect input warning indicates if test lead is plugged into current jack, but Function/Range switch isn't set to make current measurement.

Safety test leads have probe tips shrouded with insulation to expose only 3.5 mm of metal for probing. Make measurements safely in dense component areas without causing shorts with long exposed test lead tips.

204X705 5XL DMM 1-5, \$40.00 ea.; 6-up, \$38.00 ea.

Compact 3.5-Digit DMM

With Convenient Component Test Features

- Data and max hold features
- Inductance capacitance and logic capability

Compact test instrument combines DMM with component checking features in a single, easy-to-carry meter. It provides logic test, capacitance, frequency and measures voltage, current, resistance and continuity. Time-saving advantages provide increased utility and greater performance. Oversized .7" high character display has annunciators that echo dial setting for even better operator accuracy.

Features data/hold/max hold, low-battery annunciator, auto-power off and continuity beeper. Has fully fused current inputs, an industry standard usually found only in more expensive meters. Input warning beepers indicate test lead is incorrectly plugged into current input jack and meter switch isn't set on a current range. Insulated test lead tips permit safe measurements in dense component areas without causing shorts.

204X730 27XT DMM 1-5, \$120.00 ea.; 6-up, \$114.00 ea.

Mini Clamp-on DMM

Durable Style with Plenty of Measurement Range

- 400-amp max ac range
- Auto power off conserves battery

The most compact Wavetek® Meterman™ clamp-around style meter is designed to test, measure and troubleshoot typical electrical installations. It has plenty of measuring range — 8 up to 400 A and 600 V ac for effective measurements in industrial applications and commercial buildings. Also measures ohms and includes 4000-count display. It is distinguished with many time-saving, Tech-Preferred™ advantages that provide greater performance in every application including continuity check, side-accessed data hold button, autoranging capability and 30-minute auto-off.

This clamp-on style is the preferred meter of choice for testing, measuring and troubleshooting typical electrical installations. It's designed for electrical professionals who demand value, durability and high quality from their test tools. Includes test leads.

204XD32 AC60A Clamp Meter 1-5, \$80.00 ea.; 6-up, \$77.00 ea.

Probe-Type DMM

Easy Access to Tight Spots

- Simple, one-handed operation
- Continuity test with beeper

Designed for easy carrying, simple use and convenient access to tight spots, Wavetek's moderately priced device is among the smallest DMMs on the market today. Use it to measure dc voltage (420 mV-300 V), ac voltage (4.2 V-300 V) and resistance to 4.2 MΩ. Model DM73B also has a continuity beeper. The 3.5-digit LED displays value, polarity, overrange, autorange and low-battery information. This lightweight, probe-type meter can be operated

easily with one hand, allowing for full concentration on measurements. The data hold button keeps readings on the display for later evaluation. Includes test lead pair, two 1.5-volt batteries, two fuses and manual.

204X748 DM73B Probe-Type DMM 1-5, \$50.00 ea.; 6-up, \$48.00 ea.
204X010 VC-11 Carrying Case \$9.00 ea.

Full-Function Field Service Meter

Built to Be Rugged and Reliable

- Low ac/dc current ranges
- Logic testing feature

This DMM may be small in size and low in cost, but it's packed with features. Full-performance 3.5-digit 15XL measures dc voltage to 1000 V, ac voltage to 750 V and fused dc/ac current to 10 A. It has a wide resistance range to 2000 MΩ, high-speed continuity beeper and diode testing. Logic test detects and displays presence of TTL pulses up to 20 MHz. The 67" high LCD digits are easy to see from a distance.

Incorrect input warning is a safety feature to alert user of possible meter misuse. Safety test leads have probe tips shrouded with insulation to expose only 3.5 mm of metal for probing. Thanks to this feature, measurements can be made safely in dense component areas without causing shorts with long exposed test lead tips.

204X716 15XL DMM 1-5, \$60.00 ea.; 6-up, \$57.00 ea.

Pocket-Sized DMM

Ideal for Field Service

- Autoranging with 4000-count digital display
- Battery test function with go/no go LED

DM9 is Wavetek® Meterman™'s high-performance, autoranging meter in a compact package. This low-cost, small-sized multi-taster offers the features and functions to make it the perfect addition to a field service tool case. Or use it as the top-of-the-line, do-it-all workshop meter. Rugged and reliable, the analog DM9 measures up to 600 V ac/dc and ac/dc current to 400 mA. It quickly tests diodes and batteries plus features continuity beeper. High-resolution, 4000-count, 3.75-digit meter features large display with superior Digi-Glo™ backlighting, live voltage, safety tester and wrong-input warning beeper.

Includes Tech-Preferred™ safety test leads and protective holster typically found only on more expensive meters and battery for a ready-to-use solution.

204X051 DM9 DMM 1-5, \$40.00 ea.; 6-up, \$38.50 ea.

Multi-Functional Clamp Meter

Measures ac Current and Voltage

- 600-amp max ac range
- Continuity check

The AC60A is the workhorse clamp-on meter for modern electrical applications. Its Amps current measuring capability provides extra range and an extra margin of safety in a more user-safe, handguard design than similarly priced units with lower capability. In addition to current measuring, this 4000-count display meter measures 600 V ac max and resistance to 40 kΩ.

This Wavetek® Meterman™ meter is distinguished with many time-saving, Tech-Preferred™ advantages that provide increased utility and greater performance in every application. They include continuity check, side-accessed data hold button, autoranging capability, overload protected voltage and 30-minute auto-off to preserve battery life.

204X040 AC60A Clamp Meter 1-5, \$100.00 ea.; 6-up, \$96.00 ea.

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